MEAN AND VARIANCE

The expected value or mean of an RV X is by definition

$$m_X = \mu_X = E(X) = \int_{-\infty}^{\infty} x f(x)\, dx .$$

Ex: If X is uniform in the interval (x_1, x_2), find $E(X)$.

$$E(X) = \frac{1}{x_2 - x_1} \int_{x_1}^{x_2} x\, dx = \frac{1}{x_2 - x_1} \cdot \frac{1}{2} x^2 \Big|_{x_1}^{x_2} = \frac{1}{2}(x_1 + x_2).$$

Discrete type: For discrete type RVs the integral can be written as a sum. Suppose that X takes the values x_i with probability p_i. In this case

$$f(x) = \sum_i p_i\, \delta(x - x_i)$$

Using the identity

$$\int_{-\infty}^{\infty} x\, \delta(x - x_i)\, dx = x_i$$

we obtain

$$E(X) = \sum_i p_i x_i \qquad p_i = P(X = x_i)$$

Ex: If X takes the values 1, 2, ..., 6 with probability 1/6, then

$$E(X) = \frac{1}{6}(1 + 2 + \dots + 6) = 3.5$$

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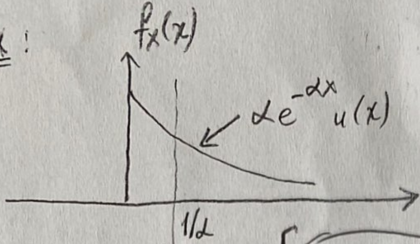
Ex:



$$m_x = \int_0^\infty x \alpha e^{-\alpha x}\, dx = \alpha \frac{e^{-\alpha x}}{(-\alpha)^2} (-\alpha x - 1) \Big|_0^\infty$$

$u = x \quad du = dx$

$dv = e^{-\alpha x} dx \quad v = -\frac{1}{\alpha} e^{-\alpha x}$

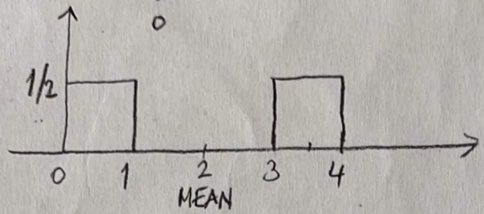
$$= \frac{\alpha}{(-\alpha)^2} \left[0 - e^0 (0-1) \right] = \frac{1}{\alpha}$$

$\alpha [uv - vdu]$

$$\alpha \left[-\frac{1}{\alpha} x e^{-\alpha x} \Big|_0^\infty + \frac{1}{\alpha} \int_0^\infty e^{-\alpha x} dx \right] = \alpha \left[-\frac{1}{\alpha^2} (0-1) \right] = \frac{1}{\alpha}$$

0

Ex:



$$m_x = \int_0^1 \frac{1}{2} x\, dx + \int_3^4 \frac{1}{2} x\, dx$$

$$= \frac{1}{4} x^2 \Big|_0^1 + \frac{1}{4} x^2 \Big|_3^4$$

$$= \frac{1}{4} + 4 - \frac{9}{4} = 2$$

Ex: Die Roll

$$F_X(x) = \frac{1}{6} \sum_{i=1}^{6} u(x-i) \qquad f_X(x) = \frac{1}{6} \sum_{i=1}^{6} \delta(x-i)$$

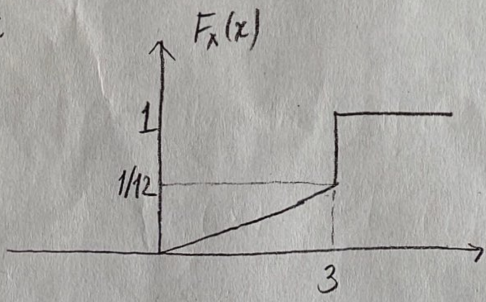
$$m_x = \int_{-\infty}^{\infty} x \frac{1}{6} \sum_{i=1}^{6} \delta(x-i) = \frac{1}{6} \left[1+2+3+4+5+6 \right] = 3{,}5$$

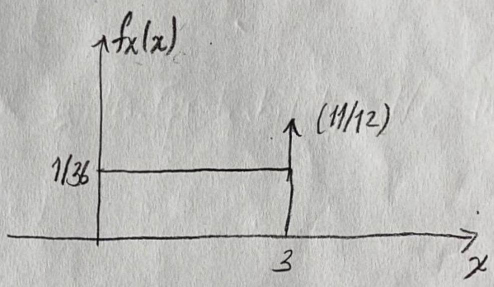
Ex: EXAMS : 65 students

$$m_x = \frac{1}{65} \sum_{i=1}^{65} x_i$$

For 12 different scores $\quad m_x = \sum_{j=1}^{12} x_i P(x_i) = 63 \cdot \left(\frac{5}{65} \right) + 98 \left(\frac{3}{65} \right)$

Ex:





$$F_X(x) = \begin{cases} 0, & x \leq 0 \\ \frac{x}{36}, & 0 \leq x < 3 \\ 1, & x > 3 \end{cases}$$

$$f_X(x) = \begin{cases} 0, & x \leq 0 \\ 1/36, & 0 \leq x < 3 \\ \frac{11}{12} \delta(x-3), & x = 3 \\ 0, & x > 3 \end{cases}$$

$$f_X(x) = \frac{1}{36} \left[u_{-1}(x) - u_{-1}(x-3) \right] + \frac{11}{12} \delta(x-3)$$

$$m_x = \int_{-\infty}^{\infty} x \left\{ \frac{1}{36}\left[u_1(x) - u_1(x-3)\right] + \frac{11}{12}\,\delta(x-3)\right\} dx$$

$$= \frac{1}{36}\,\frac{x^2}{2}\Big|_0^3 + \frac{11}{12}\cdot 3 = \frac{69}{24}$$

$\delta(t)$ – Unit Area
$= \infty$ at $t=0$
0 otherwise

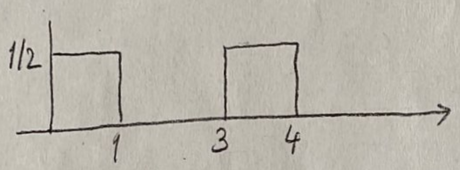
$$\int_{-\infty}^{\infty} g(x)\,\delta(x-5)\,dx = g(5)$$

n^{th} moment of $X = E\{X^n\} = \int_{-\infty}^{\infty} x^n f_X(x)\,dx$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f_X(x) = \frac{1}{2}\int_0^1 x^2\,dx + \frac{1}{2}\int_3^4 x^2\,dx$$

$$= \frac{1}{6}x^3\Big|_0^1 + \frac{1}{6}x^3\Big|_3^4$$

$$= \frac{1}{6} + \frac{64}{6} - \frac{27}{6} = \frac{19}{3}$$



$m_1 = 2$

$m_2 = 19/3, \quad \mu_2 = \sigma^2 = \frac{7}{3}$

n^{th} central moment : $\mu_n = E\{(x-m_1)^n\}$

$$\mu_1 = E[x - m_1] = E(x) - E(m_1) = m_1 - m_1 = 0$$

2^{nd} central moment = variance $\mu_2 = E\{(x-m_1)^2\} \equiv \sigma_x^2$

$$= \int_{-\infty}^{\infty} (x-2)^2 \left\{ \frac{1}{2}\left[u(x) - u(x-1)\right] + \frac{1}{2}\left[u(x-3) - u(x-4)\right]\right\} dx$$

$$= \frac{1}{2}\,\frac{(x-2)^3}{3}\Big|_0^1 + \frac{1}{2}\,\frac{(x-2)^3}{3}\Big|_3^4 \qquad \sigma^2 = \mu_2 = \frac{1}{6}\cdot 14 = \frac{7}{3}$$

Second way:

$$E[(x-m)^2] = E\{x^2 - 2mx + m^2\} = E\{x^2\} + E\{-2mx\} + E\{m^2\}$$

$$= E\{x^2\} - 2m\,\underbrace{E\{x\}}_{m} + m^2\,\underbrace{E\{1\}}_{}$$

$$E\{g(x)\} = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx \qquad = E\{x^2\} - 2m^2 + m^2\cdot 1$$

$$\boxed{\sigma^2 = E\{x^2\} - m^2}$$

$$\sigma^2 = m_2 - m_1^2 = \frac{19}{3} - 2^2 = \frac{7}{3}$$

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Mean of $g(x)$: Given an RV X and a function $g(x)$, we form the RV $Y = g(X)$. The mean of this RV is given by

$$E(Y) = \int_{-\infty}^{\infty} y f(y)\, dy.$$

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx.$$

Conditional mean: The conditional mean of a RV X assuming an event M is given by

$$E(X|M) = \int_{-\infty}^{\infty} x f(x|M)\, dx.$$

Ex.: Assume $M = \{x \leq a\}$.

$$f(x \mid X \leq a) = \frac{f(x)}{F(a)} = \frac{f(x)}{\int_{-\infty}^{a} f(x)\, dx} \quad \text{for } x < a.$$

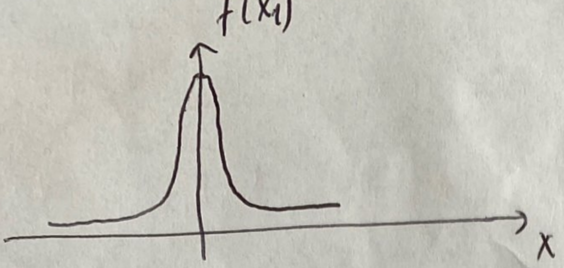
$$E(X \mid X \leq b) = \frac{\int_{-\infty}^{a} x f_X(x)\, dx}{\int_{-\infty}^{b} f(x)\, dx}$$

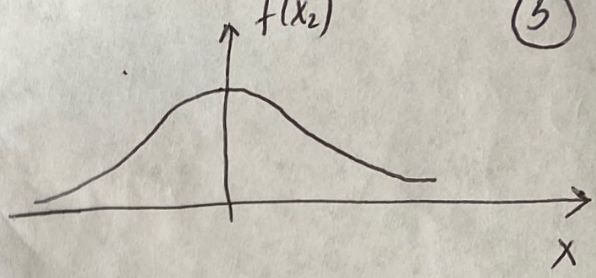
Properties: For an RV X, with expected value $\mu = E(X)$

a) $E(X - \mu) = E(X) - \mu = 0$

b) $E(aX + b) = aE(X) + b$.

Variance: Mean alone will not be able to truly represent the pdf of any RV. Consider two Gaussian RVs $X_1 : N(0,1)$ and $X_2 : N(0,3)$. Both of them have the same mean $(\mu = 0)$.





X_1 is more concentrated around the mean, whereas X_2 has a wider spread.

For an RV X with mean μ, $x-\mu$ represents the deviation of the RV from its mean. Since this deviation can be either positive or negative, consider the quantity $(x-\mu)^2$, and its average value $E((x-\mu)^2)$ represents the average square deviation of X around its mean.

$$\sigma_X^2 = E[(x-\mu)^2] \quad \text{variance. (VAR)}$$

$$\sigma_X^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx.$$

The positive square root $\sigma_X = \sqrt{E(x-\mu)^2}$ is the standard deviation.

$$VAR(X) = \sigma^2 = E[(X-\mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

$$\mu = E[X] \Rightarrow \boxed{VAR(X) = E[X^2] - E^2[X]}$$

Properties :

1. $VAR(a) = E(a^2) - E^2(a) = 0$; a is a constant.
2. $VAR(X+a) = VAR(X)$
3. $VAR(aX) = E(a^2X^2) - a^2E^2(X) = a^2\, VAR(X)$.

Ex: Find $E[X]$ and $VAR[X]$ if X is uniform in the interval $(-c, c)$.

$$f(x) = \frac{1}{2c} \qquad -c < x < c$$

$$= 0 \qquad \text{elsewhere}$$

$$E[x] = \int_{-c}^{c} x \cdot \frac{1}{2c}\, dx = \frac{1}{4c} x^2 \Big|_{-c}^{c} = \frac{1}{4c}\left(c^2 - c^2\right) = 0$$

$$VAR[X] = E[X^2] - \cancel{E^2[X]}^{\to 0}$$

$$= \int_{-c}^{c} x^2 \cdot \frac{1}{2c}\, dx = \frac{1}{6c} x^3 \Big|_{-c}^{c} = \frac{1}{6c} 2c^3 = \frac{c^2}{3}.$$

Ex.: The density of a normal RV is given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)/2\sigma^2}$$

Here, μ is the mean and σ^2 ~~are~~ is the variance of X.

Ex.: The RV X takes the values 1 and 0 with probabilities p and $q = 1-p$, respectively. Find $E[X]$ and $VAR[X]$.

$$E[X] = 1 \times p + 0 \times q = p$$

$$E[X^2] = 1^2 \times p + 0^2 \times q = p$$

$$\sigma^2 = E[X^2] - E^2[X] = p - p^2 = pq$$

* A Poisson distributed RV with parameter λ takes the values $0, 1, \ldots$ with probabilities $P(X=k) = e^{-\lambda} \dfrac{\lambda^k}{k!}$

$$E(X) = \lambda \qquad E(X^2) = \lambda^2 + \lambda \qquad \sigma^2 = \lambda.$$

Ex: The RV X is $\overset{\mu\ \sigma^2}{N(5,2)}$ and $Y = 2X+4$. Find $E(Y)$, σ_y and $f_Y(y)$.

$$E(Y) = 2E(X) + 4 = 2\times 5 + 4 = 14$$

$$VAR(Y) = 4\,VAR(X) = 4 \times \cancel{4}\ 2 = \cancel{16}\ 8 \qquad \sigma_y = \cancel{2\sqrt{2}}\ \cancel{4}\ 2\sqrt{2}$$

$$f(x) = \frac{1}{\sqrt{2\pi\,2}}\, e^{-(x-5)^2/(2\times 2)}$$

$$\left.\begin{array}{l} y = 2x+4 \Rightarrow x_1 = \dfrac{y-4}{2} \\ g'(x) = 2 \end{array}\right\} \quad f_Y(y) = \frac{f(x_1)}{|g'(x_1)|} = \frac{\frac{1}{\sqrt{4\pi}}\, e^{-\left(\frac{y-4}{2}-5\right)^2/4}}{2}$$

Ex: If X is $N(0,4)$ and $y = 3X^2$, find $E(Y)$, σ_y and $f_Y(y)$.

$$E(Y) = 3E(X^2) \qquad VAR(X) = E(X^2) - \cancel{E^2(X)}^{\,0} \Rightarrow E(X^2) = 4$$

$$E(Y) = 12$$

Hint: For a RV $X : N(0,\sigma^2) \Rightarrow E(X^4) = 3\sigma_x^4 = 3E^2(X^2)$.

$$\begin{aligned} VAR(Y) = VAR(3X^2) &= E(9X^4) - E^2(3X^2) \\ &= 9\underbrace{E(X^4)}_{48} - 9\underbrace{E^2(X^2)}_{16} \end{aligned} \quad \Bigg\} \quad \begin{array}{l} VAR(Y) = 288 \\ \sigma_y = \sqrt{288} \end{array}$$

$$E(X^4) = 3\sigma_x^4 = 48$$

$y \geq 0$:

$$x_1 = \sqrt{\frac{y}{3}} \qquad g'(x) = 6x \qquad g'(x_1) = 6\sqrt{\frac{y}{3}}$$

$$x_2 = -\sqrt{\frac{y}{3}} \qquad\qquad g'(x_2) = -6\sqrt{\frac{y}{3}}$$

$$f(y) = \frac{f_X\left(\sqrt{\frac{y}{3}}\right)}{6\sqrt{\frac{y}{3}}} + \frac{f_X\left(-\sqrt{\frac{y}{3}}\right)}{6\sqrt{\frac{y}{3}}} = \frac{2}{6\sqrt{\frac{y}{3}}}\, f_X\left(\sqrt{\frac{y}{3}}\right)$$

$$= \frac{2}{\sqrt{12y}} \cdot \frac{1}{\sqrt{8\pi}}\, e^{-\left(\sqrt{\frac{y}{3}}\right)^2/8}$$

Ex: If X is uniform in the interval $(10, 12)$ and $Y = X^3$, find $f_Y(y)$ and $E(Y)$.

$x_1 = y^{1/3}$ $\quad g'(x) = 3x^2$ $\quad g'(x_1) = 3y^{2/3}$

$f_X(x) = \frac{1}{2}, \quad 10 < x < 12$

$f_Y(y) = \frac{f_X(y^{1/3})}{3y^{2/3}} = \frac{1}{6} \cdot \frac{1}{y^{2/3}}, \quad 10^3 < y < 12^3$.

$E(Y) = E(X^3) = \int_{10}^{12} x^3 \cdot \frac{1}{2}\, dx = 1342.$

Ex: Find $E(Y)$ and σ_Y if $Y = (X - \mu_X)/\sigma_X$.

$E(Y) = \frac{E(X - \mu_X)}{\sigma_X} = 0$

$VAR(Y) = E(Y^2) - \cancel{E^2(Y)}^{\,0} = \frac{E[(X-\mu_X)^2]}{\sigma_X^2} = \frac{\sigma_X^2}{\sigma_X^2} = 1.$